Core Use Cases Sprint 1

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UC-01 Authentication and user management

Related stories: CORE-01 Registration and login, CORE-02 User profiles, CORE-03 Role management.

Actor: A borrower, broker or bank official.

Purpose: Let users access AL with an account, saved profile and appropriate role.

Preconditions: Firebase Authentication and Firestore are available. Staff roles must be assigned through an authorised process.

Trigger: The user chooses to register or log in.

Main flow

1. The user chooses to register, and the system displays the registration form.
2. The user enters and submits their credentials and profile details. The system validates the details. Firebase Authentication creates the account and returns its unique user ID. The system saves the profile and permitted role in Firestore, linked to that ID.
3. The user logs in with valid credentials. The system authenticates the user through Firebase Authentication.
4. The user opens their account. The system retrieves the saved profile and role, then allows access according to that role.

Alternative flows and errors

- Existing account: the user starts at the login step.

- Invalid details or duplicate account: registration fails and the user can correct the details or log in. Invalid login credentials do not grant access.

- Missing profile or failed profile save: the system reports the problem and blocks access that needs a role. The profile is recovered using the existing user ID.

- Unavailable service or unknown role: the system reports the error and refuses the requested access. Users cannot gain staff access by changing a submitted role value.

Outcome

The user is authenticated, the profile is linked to the correct user ID, and access follows the stored role. Failed attempts do not grant protected access. If account creation succeeds but the profile save fails, profile recovery is still required.

UC-02 Mortgage application data

Related stories: CORE-04 Application storage, CORE-05 Financial details, CORE-06 Property details.

Actor: A borrower.

Purpose: Save a mortgage application with its financial and property details so it can be opened later.

Preconditions: The borrower is logged in and has a saved profile. Required fields and validation rules are defined.

Trigger: The borrower chooses to create a mortgage application.

Main flow

1. The borrower opens the application form, and the system displays the required application fields.
2. The borrower enters their income, existing debt, deposit and requested mortgage term.
3. The borrower enters the property location or county, area, bedrooms, property type and BER rating.
4. The borrower submits the application. The system checks their identity and validates the fields. It creates a unique application ID and saves the application in Firestore, linked to the borrower.
5. After the save succeeds, the system confirms the submission and returns the application ID. The financial and property details are linked to that application.
6. The borrower opens the saved application later. The system checks access, retrieves the application and displays its saved details.

Alternative flows and errors

- Missing or invalid fields: the system identifies what needs correcting and asks the borrower to submit again.
- Login expires: the borrower must log in before continuing.
- Save fails: the system reports the failure and does not show a success confirmation. The borrower can retry once the problem is resolved.
- Application not found: the system explains that the requested application is unavailable.

- Unauthorised request: access is refused. Borrowers can only access their own applications.

Outcome

A successfully saved application has a unique ID and can be retrieved from Firebase. Its financial and property details are linked to the correct record. Invalid input is rejected before saving, and an unconfirmed save is reported as a failure.

UC-03 Application tracking and broker assignment

Related stories: CORE-07 Application status, CORE-08 Broker assignment.

Actor: A staff member with permission to update the application.

Purpose: Keep the application status up to date and record its assigned broker.

Preconditions: The staff member is logged in and has the required access. The application exists. Assignment needs an existing broker profile. Status changes follow the lifecycle rules.

Trigger: The staff member opens an application to review or update it.

Main flow

1. The staff member selects an application. The system checks access and retrieves the application, its current status and any assigned broker.
2. The staff member requests a status change. The system checks permission and the lifecycle rules, then saves the permitted status change.
3. The staff member selects a broker. The system checks assignment permission and confirms that the selected profile has the broker role.
4. The staff member saves the assignment. The system stores the broker ID against the application and confirms success.
5. The staff member opens the application again. The system retrieves and displays the saved status and assigned broker.

Alternative flows and errors

- View only: the actor can read the details without changing them. Borrowers can view the status and assigned broker for their own application.

- One change only: the actor can update the status or the broker assignment separately. The other value stays unchanged.

- Invalid status or disallowed transition: the system rejects the change and keeps the previous status.

- Missing broker or wrong role: the system rejects the assignment and keeps any previous assignment.

- No permission, missing application or failed save: the system reports the problem and does not confirm success. A failed assignment does not undo an earlier saved status change.

Outcome

Successful changes remain saved against the correct application and can be retrieved later. Status values include Applied, AI Deliberated, Underwriter Approved, Funded and Closed.